

Normalizing features

Evert Jan Karman

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In this course, normalizing features specifically indicates: scaling each feature column by the L2 norm of the feature column. So for each feature h_j calculate:

$$Z_j = \sqrt{\sum_{i=0}^{N-1} h_j(x_i)^2}$$

And then:

$$\forall k \quad \underline{h}_j(x_k) = \frac{h_j(x_k)}{Z_j}$$

Note that this happens on the training data. Later, the test data must be scaled by the same factor Z_j , that was obtained using the training data.